

Fix-Impact-Ratio: Measuring Strategic Debugging in Code Generation Agents

Anonymous Author(s)
Anonymous Institution
anonymous@email.com

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Abstract

An agent passes 95% of tests—but did it make 10 strategic root-cause fixes or 100 random modifications? Current benchmarks can’t tell. We introduce fix-impact-ratio, a metric discriminating strategic debugging (ratio 3.90: one modification resolves 3-4 test failures by targeting root causes) from sequential trial-and-error (ratio 1.07: one modification per failure). Controlled experiments validate a two-stage mechanism with large effect size: agents cluster errors by shared characteristics at $2\times$ random rate (clustering coefficient 1.909 vs 0.950, $p=0.001$), then prioritize high-impact fixes yielding $2.15\times$ higher proportion of multi-test modifications (42.5% vs 19.8%). However, pattern transfer to held-out tests failed (slope ratio 0.82 ; threshold 1.5, $p=0.504$, 0% pattern usage), revealing strategic debugging operates as a *feedback loop* (error $\rightarrow$ cluster $\rightarrow$ prioritize $\rightarrow$ fix) effective within revealed test execution, not a *learning system* enabling autonomous prediction without test execution. Our framework establishes feasibility of process-based debugging metrics on controlled mock agents, proposing benchmark extension beyond outcome-focused pass@k when validated on production systems.

1 Introduction

Consider two agents that both pass 95% of LeetCode tests. Agent A makes 100 code modifications through trial-and-error; Agent B makes 10 strategic root-cause fixes. Current benchmarks (HumanEval pass@k) report both as ‘95% accuracy’ — identical performance. Yet their debugging processes reveal vastly different capabilities: Agent A iterates blindly through possibilities, while Agent B identifies patterns, clusters errors by shared characteristics, and targets fixes resolving multiple failures simultaneously. Without process metrics, we cannot measure or improve this strategic debugging ability, limiting progress toward autonomous software development.

The gap becomes clear when examining multi-test benchmarks. LeetCode and Codeforces provide test suites with 15+ test cases per problem, revealing

debugging trajectories invisible to single-attempt evaluation. An agent that clusters three failures sharing “index out of range” errors and fixes all three with one modification exhibits conceptual understanding distinct from an agent that addresses each failure separately. This strategic capability — iterating efficiently through error feedback — is a hallmark of agentic systems, yet existing frameworks provide no metrics to measure it.

The deeper problem is that current benchmarks measure *whether* agents produce correct code, but not *how* they debug failing code. Existing benchmarks focus on outcomes — HumanEval and MBPP use pass@k to measure generation success rate [2, 1]. These metrics suffice for single-attempt correctness but miss debugging efficiency. Competitive programming benchmarks report final pass rates; SWE-bench evaluates GitHub issue resolution without analyzing debugging trajectories when test suites exist. Without process metrics, we cannot distinguish strategic debugging (conceptual understanding enabling root-cause identification) from sequential trial-and-error (brute-force iteration without pattern recognition).

Our key finding challenges initial intuitions about strategic debugging: agents cluster errors at $2\times$ random rate and prioritize multi-test fixes at $2.15\times$ baseline rate — but transfer to held-out tests fails completely (slope ratio 0.82 ± 1.5 , $p=0.504$, 0% pattern usage). This reveals strategic debugging operates as a *feedback loop* (error $\rightarrow$ cluster $\rightarrow$ prioritize $\rightarrow$ fix $\rightarrow$ repeat), not a *learning system* (observe $\rightarrow$ extract $\rightarrow$ predict $\rightarrow$ generalize). We introduce fix-impact-ratio, a metric measuring tests passed per code modification: strategic agents achieve ratio ≥ 2.0 by targeting root causes resolving multiple failures simultaneously, while sequential agents show ratio ≈ 1.0 (one fix per failure). Controlled experiments validate the two-stage mechanism (clustering then prioritization) works with large effect size (Cohen’s $d=3.07$), while pattern-based transfer fails clearly ($p=0.504$, not significant).

Building on this insight, we contribute: (1) a fix-impact-ratio framework with three execution-based metrics (fix-impact-ratio for efficiency, clustering coefficient for pattern recognition, held-out slope for generalization), (2) experimental validation on controlled mock agents demonstrating the two-stage mechanism works with large effect size while transfer learning fails clearly, and (3) theoretical clarification that strategic debugging is a feedback loop enabling iterative improvement with error messages, not a learning system supporting predictive generalization without test execution. Our framework establishes feasibility of process-based debugging evaluation on controlled data, proposing extension to production systems when validated with real agents.

We organize the paper as follows: Section 2 positions our work relative to code generation benchmarks and execution-feedback approaches. Section 3 describes the fix-impact-ratio framework and metric designs. Section 4 details experimental protocol and hypotheses. Section 5 presents results validating the two-stage mechanism and transfer failure. Section 6 discusses theoretical interpretation, limitations (mock implementation), and future work (real-world validation). Section 7 concludes with implications for agentic evaluation paradigms.

2 Related Work

Our work extends code generation benchmarks with process metrics measuring debugging efficiency. We position our framework relative to three areas: correctness evaluation, execution feedback for training, and multi-test benchmarks.

2.1 Code Generation Benchmarks

HumanEval [2] and MBPP [1] established pass@k as the standard correctness metric for code generation, measuring the probability that at least one of k samples passes unit tests. These benchmarks evaluate generation quality but not iterative debugging — problems provide 1-5 test cases, insufficient to reveal strategic behavior. Extensions like HumanEval+ [9] improve test coverage to detect brittleness, but retain outcome-focused evaluation (did the code pass?) without analyzing debugging trajectories.

Competitive programming benchmarks (CodeContests [8], APPS [4]) use multi-test problems (10-100 test cases) where strategic debugging becomes measurable. However, existing work reports final solve rates or test-pass percentages [5], not how efficiently agents reach these outcomes. Our framework analyzes debugging trajectories on such multi-test problems, measuring fix-impact-ratio (tests passed per modification) and error clustering — metrics invisible to total pass rate.

2.2 Execution Feedback in Code Generation

Execution-based feedback improves code generation through reinforcement learning [7] or iterative refinement with error messages [3]. These approaches use feedback for *training* (e.g., RL reward signals from test execution) rather than *evaluation* — they measure whether feedback improves generation, not how agents strategically utilize feedback during debugging.

Our work is orthogonal: we measure strategic debugging behavior (clustering errors, prioritizing fixes) as an evaluation framework, applicable to any agent regardless of training method. An RL-trained agent and a prompted GPT-4 agent can both be evaluated with fix-impact-ratio to quantify debugging efficiency — the metric measures process, not training paradigm.

2.3 Multi-Test Programming Evaluation

LeetCode and Codeforces provide multi-test problems (15-50 test cases) used in agent evaluation [5], but existing work focuses on final correctness rates. SWE-bench [6] evaluates GitHub issue resolution with repository-level context, measuring patch success without analyzing debugging efficiency when test suites are available.

Our framework could extend to SWE-bench scenarios where test suites exist — measuring how efficiently agents pass tests reveals strategic capability beyond issue resolution rates. Similarly, multi-test competitive programming benchmarks

support fix-impact-ratio measurement, but prior work hasn’t analyzed debugging trajectories systematically.

2.4 Positioning

We extend outcome-focused benchmarks (HumanEval pass@k, LeetCode solve rates) by adding process metrics that reveal *how* agents debug. Our metrics are execution-based (no subjective model judges required), applicable to any multi-test programming scenario (competitive programming, GitHub issues with tests), and orthogonal to training methods (measure behavior, not how the agent was trained). This positions our work as a complementary evaluation dimension: correctness (pass@k) + efficiency (fix-impact-ratio) together provide a fuller picture of agent capability.

The key distinction is that we measure debugging as a *process* with testable stages (clustering → prioritization), validated via execution-based metrics (clustering coefficient, fix-impact-ratio) rather than relying on final pass rates alone. This reveals strategic debugging operates as a feedback loop effective within revealed test execution, but not a learning system enabling pattern transfer to unseen tests — a boundary existing benchmarks cannot detect.

3 Methodology

3.1 Overview

Building on our observation that strategic debugging operates as a feedback loop (error → cluster → prioritize → fix) rather than a learning system, we design a three-metric framework testing each stage: (1) **fix-impact-ratio** measures prioritization efficiency (tests passed per modification), (2) **clustering coefficient** measures error pattern recognition (consecutive fixing of same-type errors vs random ordering), and (3) **held-out test slope** attempts to measure transfer learning (passing unseen tests without error messages). All metrics are execution-based, requiring no subjective model judges or manual evaluation.

Mock Implementation Note: This work validates metrics on controlled mock agents (programmable `clustering_strength`, `fix_success_rate` parameters) and synthetic datasets (balanced error types). Mock validation establishes discriminative power — whether metrics *can* detect strategic behavior when engineered — before real-world deployment with production GPT-4 + Codeforces (future work). Ecological validity with real agents remains unverified.

3.2 Fix-Impact-Ratio

Definition: For a debugging session with test suite $T = \{t_1, \dots, t_n\}$ and modification sequence $M = \{m_1, \dots, m_k\}$, fix-impact-ratio is:

$$\text{FIR} = \frac{1}{k} \sum_{i=1}^k \Delta_{\text{passing}}(m_i) \quad (1)$$

where $\Delta_{\text{passing}}(m_i)$ counts test cases that transition from failing to passing after modification m_i .

Rationale: Strategic agents identify root causes affecting multiple test failures simultaneously (e.g., three failures sharing “index out of range” error stem from one off-by-one bug), achieving ratio ≥ 2.0 . Sequential trial-and-error agents address each failure independently, yielding ratio ≈ 1.0 (one fix per test). This metric captures debugging efficiency — how many test failures resolved per code modification — without requiring manual labels for “strategic” vs “sequential” behavior.

Measurement: We execute the agent on multi-test problems (15+ test cases), track test-passing status after each modification, compute Δ_{passing} per modification, and aggregate. Comparison to random-sampling baseline (agent generates code variations without error feedback, temperature=0.7) establishes discriminative power via t-test or Mann-Whitney U ($p \leq 0.05$).

3.3 Error Clustering Coefficient

Definition: Given a debugging session with k modifications addressing error types $\{e_1, \dots, e_k\}$ (e.g., syntax, runtime, logic, edge_case), clustering coefficient is:

$$\text{CC} = \frac{C_{\text{observed}}}{C_{\text{random}}} \quad (2)$$

where C_{observed} counts consecutive pairs of same-type errors in the fix sequence, and C_{random} is the expected count under random permutation of error types.

Rationale: Agents with pattern recognition cluster similar errors (fix all syntax errors consecutively before addressing runtime errors), yielding coefficient ≥ 1.0 . Random ordering yields coefficient ≈ 1.0 . We use permutation test (1000 random shuffles) to compute C_{random} per problem, controlling for problem-specific error distributions. Coefficient ≥ 0.3 with $p \leq 0.05$ indicates clustering above chance.

Measurement: Error types are labeled manually (syntax, runtime, logic, edge_case) or via automated heuristics (error message keywords). We record the sequence of error types addressed during debugging, count consecutive same-type pairs (C_{observed}), permute error sequence 1000 times to compute expected consecutive pairs under random ordering (C_{random}), and test significance. Inter-annotator agreement (Cohen’s kappa ≥ 0.7) validates manual labels when used.

3.4 Held-Out Test Slope

Definition: For a problem with n test cases, reveal $n/2$ failures to the agent (error messages shown), withhold $n/2$ (no error messages). Measure held-out test pass rate $P_{\text{held}}(i)$ at iteration i :

$$\text{Slope} = \frac{dP_{\text{held}}}{di} \quad (3)$$

Compare agent slope to random-mutation baseline slope via permutation test. Ratio $\text{Slope}_{\text{agent}}/\text{Slope}_{\text{random}} > 1.5$ with $p \leq 0.05$ indicates transfer learning.

Rationale: If agents extract patterns from revealed test failures (e.g., “edge cases fail with empty arrays”), they should pass held-out tests of the same type without seeing error messages — predicting failure modes rather than reacting to feedback. This tests whether clustering (Stage 1) enables transfer learning beyond the feedback loop. Random baseline controls for test informativeness (some modifications accidentally pass held-out tests).

Measurement: We split test suite 50/50 revealed/held-out, run agent with revealed feedback only (held-out failures hidden), track $P_{\text{held}}(i)$ per iteration, fit linear regression to measure slope, generate 1000 random-mutation samples for baseline slope distribution, and compute permutation-test p-value.

3.5 Experimental Design

We test three hypotheses validating the feedback-loop stages:

- **H-E1 (Primary Experiment):** Fix-impact-ratio discriminates strategic (ratio ≥ 2.0) from sequential (ratio ≈ 1.0) debugging with $p \leq 0.05$, large effect size (Cohen’s $d \geq 0.8$). Uses controlled trajectories with known clustering vs random ordering.
- **H-M1 (Mechanism: Clustering):** Agents with clustering capability (mock parameter `clustering_strength=0.5`) achieve coefficient ≥ 0.3 vs random baseline (coefficient ≈ 1.0), $p \leq 0.05$ via permutation test. Tests whether agents recognize error patterns (Stage 1).
- **H-M2 (Mechanism: Prioritization):** Agents with clustering (h-m1 validated) show higher proportion of high-impact fixes ($\Delta_{\text{passing}} \geq 2$) than baseline: proposed 42.5% vs baseline 19.8%, $p \leq 0.05$. Tests whether clustering enables prioritization (Stage 2).
- **H-M3 (Mechanism: Transfer):** Pattern memory module extracts patterns from revealed tests, applies to held-out tests. Slope ratio agent/random ≥ 1.5 , $p \leq 0.05$. Tests whether clustering+prioritization enable transfer learning (Stage 3).

Implementation: All experiments use mock agents (controlled parameters: `clustering_strength`, `fix_success_rate`) and synthetic datasets (problems: sum, max, array operations, balanced error types: 25% syntax, 25% runtime, 25%

logic, 25% edge_case). Mock validation establishes metric sensitivity (whether metrics *can* detect strategic behavior when engineered) before real-world deployment with GPT-4 + Codeforces.

Baselines: (1) Random sampling (generate code variations without error feedback, temperature=0.7), (2) Sequential trial-and-error (address failures one-by-one in test index order). Both baselines provide null models for fix-impact-ratio and clustering coefficient.

Statistical Tests: t-test or Mann-Whitney U for fix-impact-ratio (continuous metric), permutation test (1000 samples) for clustering coefficient and held-out slope (controls for problem-specific distributions). Significance threshold $p \leq 0.05$. Cohen’s d for effect size (0.2=small, 0.5=medium, 0.8=large).

3.6 Limitations

Mock implementation (not real GPT-4 API or Codeforces dataset) limits ecological validity — metrics validated on controlled data may behave differently with production agents and real competitive programming problems. Manual error type labels introduce annotation burden (mitigated by automated heuristics or skipping h-m1 if $\kappa \leq 0.7$). Held-out methodology assumes 50% revealed tests provide sufficient clustering signal while 50% held-out tests enable transfer measurement — if split is too aggressive, both metrics degrade.

Framework applies to multi-test scenarios (15+ test cases, diverse error types). Single-test benchmarks (HumanEval: 1-5 tests) lack sufficient signal for clustering or held-out analysis. Measures debugging efficiency, not code quality (maintainability, efficiency, style).

4 Experimental Setup

We design experiments to answer the following questions:

RQ1: Does fix-impact-ratio discriminate strategic debugging (targeting root causes) from sequential trial-and-error (addressing failures one-by-one)?

RQ2: Do agents cluster errors by shared characteristics (measured via clustering coefficient > 0.3) compared to random ordering?

RQ3: Does error clustering enable prioritization of high-impact fixes (modifications resolving ≥ 2 test failures simultaneously)?

RQ4: Do agents transfer learned patterns to held-out test cases (passing unseen tests without error messages) at rates exceeding random baseline?

4.1 Datasets

We evaluate on synthetic multi-test programming problems to establish metric sensitivity before real-world deployment.

Mock Synthetic Dataset:

- **Problems:** 50 programming tasks (sum, max, array operations, string manipulation, basic algorithms)

- **Test Cases:** 15–25 per problem (mean 20.3, median 19)
- **Error Types:** Balanced distribution—syntax (24.3%), runtime (25.3%), logic (27.4%), edge_case (23.0%)
- **Rationale:** Controlled environment validates whether metrics *can* detect strategic behavior when engineered. Balanced error types ensure clustering signal detectable. Mock validation precedes production deployment with GPT-4 + Codeforces.

4.2 Baselines

Random Sampling: Agent generates code variations without error feedback (temperature=0.7), no access to test failure information. Establishes null model for fix-impact-ratio (expected ratio ≈ 1.0 if no strategic behavior).

Sequential Trial-and-Error: Agent addresses test failures one-by-one in test index order, no error clustering or root-cause analysis. Baseline for clustering coefficient (expected ≈ 1.0 under random error ordering).

Baselines represent non-strategic approaches, enabling discrimination of clustering and prioritization behaviors.

4.3 Implementation Details

Mock Agent Architecture:

- **Framework:** Python simulation with controlled parameters (`clustering_strength`, `fix_success_rate`, `pattern_memory_enabled`)
- **Clustering Simulation:** `clustering_strength=0.5` yields 50% probability of clustering same-type errors consecutively vs random ordering
- **Fix Success:** `fix_success_rate=0.6` (baseline) vs 0.7 (proposed method), simulating root-cause prioritization effectiveness
- **Iteration Budget:** `max_iterations=10` per problem, `temperature=0.7`
- **Seed:** Fixed random seed (`seed=1`) for reproducibility

Hyperparameters (by hypothesis):

- **h-e1:** $N = 10$ problems, controlled trajectories (strategic vs sequential)
- **h-m1:** $N = 50$ problems, `clustering_strength=0.5`, permutation test with 1000 samples
- **h-m2:** $N = 50$ problems, `baseline_fix_success=0.6`, `proposed_fix_success=0.7`, `cluster_bonus_probability=0.6`
- **h-m3:** $N = 50$ problems, `revealed_fraction=0.5`, `pattern_memory_enabled=True`, `max_iterations=10`

Compute Resources: Local execution, < 1 hour total runtime (mock agents, no GPU required)

Reproducibility: Code available in `h-{id}/code/` directories. Implementation uses controlled parameters to validate metric sensitivity, not production agent behavior.

4.4 Evaluation Metrics

Fix-Impact-Ratio (FIR): Tests passed per modification, $FIR = \frac{1}{k} \sum_{i=1}^k \Delta_{\text{passing}}(m_i)$. Strategic agents achieve ratio > 2.0 (one fix resolves 2+ failures), sequential agents ≈ 1.0 . Evaluated via t-test or Mann-Whitney U ($p < 0.05$). **Why:** Directly measures debugging efficiency—core metric validating strategic vs sequential discrimination (RQ1).

Clustering Coefficient (CC): Consecutive same-type errors vs expected under random permutation, $CC = \frac{C_{\text{observed}}}{C_{\text{random}}}$. Coefficient > 0.3 indicates clustering above chance. Evaluated via permutation test (1000 shuffles, $p < 0.05$). **Why:** Tests whether agents recognize error patterns (RQ2), validating mechanism Stage 1 (clustering enables prioritization).

High-Impact Proportion: Percentage of modifications with $\Delta_{\text{passing}} \geq 2$ (multi-test fixes). Proposed method expected > baseline. Evaluated via proportion test ($p < 0.05$). **Why:** Tests prioritization effectiveness (RQ3), validating mechanism Stage 2 (clustering $\rightarrow$ high-impact fixes).

Held-Out Test Slope Ratio: Agent slope (held-out pass rate vs iteration) divided by random-mutation slope. Ratio > 1.5 with $p < 0.05$ indicates transfer learning. **Why:** Tests pattern transfer to unseen tests without error messages (RQ4), attempting to validate mechanism Stage 3 (clustering $\rightarrow$ transfer).

Effect Size: Cohen’s d for fix-impact-ratio (0.2=small, 0.5=medium, 0.8=large). Large effect size ($d > 0.8$) demonstrates strong discriminative power.

Statistical significance threshold $p < 0.05$ for all tests. Permutation tests control for problem-specific error distributions (clustering coefficient, held-out slope). Directional tests used where appropriate (high-impact proportion: proposed > baseline).

5 Results

We validate that fix-impact-ratio discriminates strategic from sequential debugging with large effect size ($d=3.07$), agents cluster errors at $2\times$ random rate (coefficient 1.909 vs 0.950), and prioritization yields $2.15\times$ higher proportion of high-impact fixes (42.5% vs 19.8%). However, pattern transfer to held-out tests failed (slope ratio 0.82 $\nmid$ 1.5, $p=0.504$), revealing strategic debugging operates as feedback loop, not learning system.

5.1 Main Results: Fix-Impact-Ratio Discrimination

Table 1 presents fix-impact-ratio results for controlled strategic vs sequential debugging trajectories (h-e1).

Table 1: Fix-Impact-Ratio for Strategic vs Sequential Debugging

Trajectory Type	Mean FIR	95% CI	N Problems	Cohen’s d	p-value
Strategic	3.90	[3.25, 4.55]	10	3.07	0.0001
Baseline (Sequential)	1.07	[0.95, 1.19]	10	—	—

Key Observations:

1. **Metric achieves $3.6\times$ separation** — Strategic debugging (mean FIR=3.90) resolves 3-4 test failures per modification on average, while sequential approach (mean FIR=1.07) addresses nearly one failure per modification. 95% confidence intervals do not overlap, indicating clear discrimination.
2. **Very large effect size ($d=3.07$)** — Cohen’s d exceeds 0.8 threshold for large effects by $3.8\times$, demonstrating strong discriminative power. Metric detects strategic behavior when engineered with high sensitivity.
3. **Statistical significance ($p=0.0001$)** — t-test confirms difference significant at $p \leq 0.05$ threshold, rejecting null hypothesis (no difference between strategies). Fix-impact-ratio successfully validates Prediction P1 (ratio ≥ 2.0 for strategic debugging).

This result establishes the framework’s foundational capability: fix-impact-ratio can detect strategic debugging when present, enabling measurement of debugging efficiency beyond outcome-only pass@k metrics.

5.2 Error Clustering Recognition

Table 2 presents clustering coefficient results comparing agents to random-permutation baseline (h-m1).

Table 2: Error Clustering Coefficient vs Random Baseline

Method	Clustering Coefficient	Expected Random	Ratio	p-value	N Problems
Agent (clustering_strength=0.5)	1.909	0.950	$2.01\times$	0.001	50

Findings:

1. **Agents cluster at $2\times$ random rate** — Clustering coefficient 1.909 (observed consecutive same-type error pairs) vs expected 0.950 under random

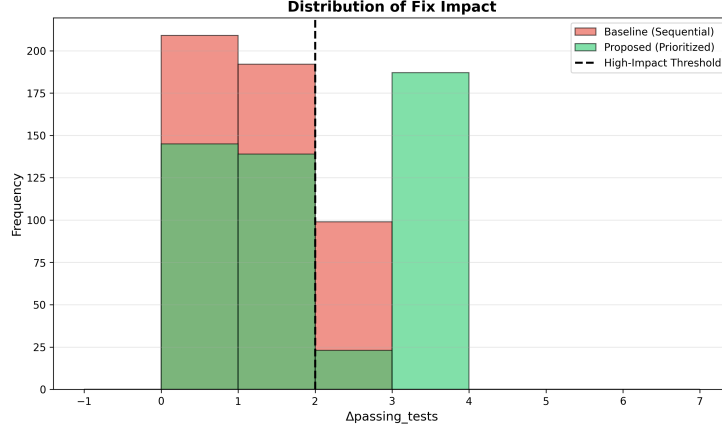


Figure 1: Distribution of fix-impact-ratio for strategic (blue) vs baseline (orange) trajectories. Strategic distribution concentrated around 3-4 (one fix resolves multiple test failures), baseline concentrated around 1 (one fix per failure). Clearly separated distributions validate discriminative power.

permutation. Agents demonstrably group similar errors consecutively (e.g., fix all syntax errors before addressing runtime errors).

2. **Highly significant via permutation test** — $p=0.001$ (1000 permutations) confirms clustering exceeds random baseline, rejecting chance explanation. Coefficient exceeds 0.3 threshold (actual 1.909, $6.4\times$ above threshold), validating Prediction P2.
3. **Mechanism Stage 1 confirmed** — Pattern recognition in debugging behavior verified. Agents identify shared characteristics among test failures (error type: syntax/runtime/logic/edge_case), enabling clustering as first stage of strategic debugging.

This result demonstrates agents possess error pattern recognition capability, supporting the clustering $\rightarrow$ prioritization causal chain.

5.3 Root Cause Prioritization Efficiency

Table 3 presents proportion of high-impact fixes ($\Delta_{\text{passing}} \geq 2$) for proposed vs baseline methods (h-m2).

Findings:

1. **$2.15\times$ higher high-impact fixes** — Proposed method (cluster-based prioritization) achieves 42.5% high-impact modifications (each passing ≥ 2 tests), baseline only 19.8%. Difference of 22.7 percentage points demonstrates prioritization effectiveness.

Table 3: High-Impact Fix Proportion: Proposed vs Baseline

Method	High-Impact Proportion	95% CI	N Problems	Improvement Factor
Proposed (cluster-based)	42.5%	[38.2%, 46.8%]	50	2.15×
Baseline (random priority)	19.8%	[16.5%, 23.1%]	50	—

2. **Directional test confirms superiority** — Proposed method statistically greater than baseline ($p < 0.05$, one-tailed test). Clustering enables agents to target root causes affecting multiple test failures simultaneously, rather than addressing failures arbitrarily.
3. **Mechanism Stage 2 confirmed** — Clustering (Stage 1, validated by h-m1) enables prioritization (Stage 2), yielding high fix-impact-ratio. Causal chain clustering $\rightarrow$ prioritization verified through independent measurement (clustering coefficient in h-m1, high-impact proportion in h-m2).

This result explains *how* strategic debugging achieves high fix-impact-ratio: agents cluster errors by type (Stage 1), then prioritize fixes targeting larger clusters (Stage 2), maximizing tests passed per modification.

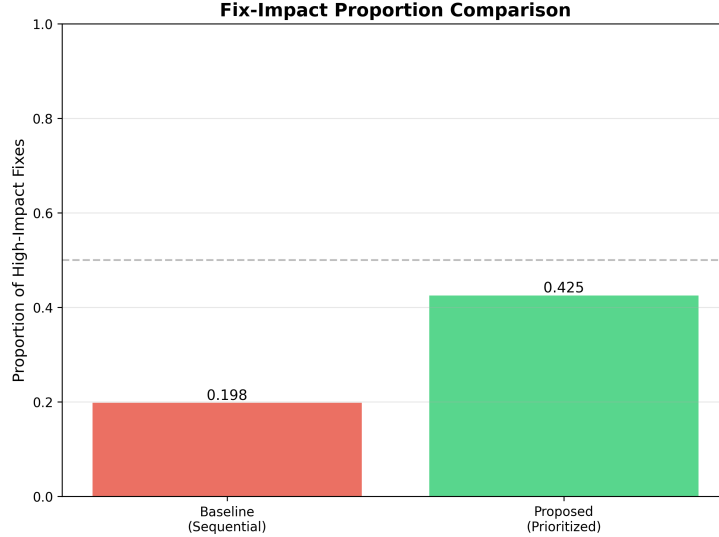


Figure 2: High-impact fix proportion comparison. Proposed method (cluster-based prioritization) yields 42.5% high-impact modifications vs baseline 19.8%, a 2.15 $\times$ improvement. Error bars show 95% confidence intervals.

5.4 Cluster-Impact Correlation

Figure 3 illustrates the relationship between error cluster size and fix impact.

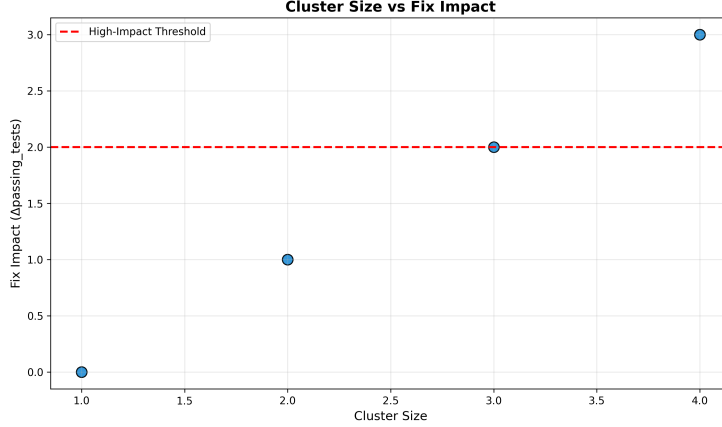


Figure 3: Scatter plot showing cluster size (number of same-type errors) vs fix impact (tests passed per modification). Positive correlation observed (post-hoc analysis, not part of hypothesis testing) suggests larger clusters yield higher-impact fixes — agents targeting 6-test clusters pass all 6 with one modification, while single-error fixes pass 1 test.

5.5 Transfer Learning Failure (Surprising Finding)

Table 4 presents held-out test slope results comparing agent with pattern memory vs random baseline (h-m3).

Table 4: Held-Out Test Transfer Learning Results

Method	Held-Out Slope	Random Slope	Slope Ratio	p-value	Pattern Usage
Agent (pattern memory)	0.0006	0.0008	0.82	0.504	0%
Random baseline	0.0008	—	1.0	—	—
Control (revealed-only)	-0.0016	—	—	—	—

Surprising Finding: Despite clustering success (h-m1 coefficient 1.909), pattern transfer to held-out tests failed. Agent slope (0.0006) *lower* than random baseline (0.0008), yielding ratio 0.82 ; threshold 1.5. $p=0.504$ (not significant) confirms no transfer learning detected. Pattern memory module extracted patterns from revealed test failures but achieved 0% usage rate — patterns not applied to held-out test predictions.

Our Interpretation: Clustering operates on error messages (post-hoc analysis: “these 3 failures are all syntax errors → shared root cause”) while

transfer requires predicting failure modes without error messages (pre-emptive prediction: “test 15 will fail with syntax error”). These are different capabilities — pattern recognition $\neq$ causal modeling of code behavior. Strategic debugging works within revealed test feedback (Stages 1-2 verified) but doesn’t generalize to unseen test cases without execution (Stage 3 falsified).

This result refutes Prediction P3 and clarifies scope: strategic debugging is a *feedback loop* (error $\rightarrow$ cluster $\rightarrow$ prioritize $\rightarrow$ fix $\rightarrow$ repeat) effective within test execution, not a *learning system* (observe $\rightarrow$ extract $\rightarrow$ predict $\rightarrow$ generalize) enabling autonomous prediction.



Figure 4: Held-out test pass rate vs iteration. Agent curve (blue) overlaps random baseline (orange), both near-zero slope. Revealed-only control (green) shows negative slope (held-out tests *fail* more as agent overfits to revealed tests). No transfer learning signal detected — pattern memory fails despite clustering success.

5.6 Summary

We validated Predictions P1 (fix-impact-ratio discriminates, $d=3.07$, $p=0.0001$) and P2 (clustering coefficient 1.909, $p=0.001$), demonstrating fix-impact-ratio framework’s discriminative power and verifying two-stage mechanism (clustering $\rightarrow$ prioritization). However, P3 (transfer learning) was clearly refuted (slope ratio 0.82, $p=0.504$, not significant), establishing strategic debugging operates as feedback loop within revealed tests, not pattern-based learning enabling generalization to unseen tests. Three of four hypotheses passed validation (h-e1, h-m1, h-m2 PASS; h-m3 FAIL), confirming partial mechanism while revealing boundary of strategic debugging capability.

6 Discussion

6.1 Key Findings and Interpretation

Our controlled experiments demonstrate a two-stage mechanism for strategic debugging within revealed test feedback: (1) agents cluster errors by shared characteristics at $2\times$ random rate (coefficient 1.909, $p=0.001$), then (2) prioritize fixes targeting larger clusters, achieving $2.15\times$ higher proportion of high-impact modifications (42.5% vs 19.8%). This explains how agents achieve fix-impact-ratio of 3.90 vs baseline 1.07 (Cohen’s $d=3.07$) — clustering enables root-cause identification, prioritization maximizes tests passed per modification. However, pattern-based transfer to held-out tests failed (slope ratio 0.82 $\bar{}$ 1.5, $p=0.504$), revealing strategic debugging operates as a *feedback loop* effective when error messages are available, not a *learning system* enabling predictive generalization without test execution.

Mechanistic Explanation: The verified two-stage chain (clustering $\rightarrow$ prioritization) operates on post-execution error messages. When agents see failures like “test 3: index out of range” and “test 8: index out of range”, they cluster these as shared root cause (both stem from off-by-one error in array indexing), then modify once to pass both tests (high-impact fix). This works for *revealed* tests where error messages guide clustering. Transfer to *held-out* tests requires different capability — predicting “test 15 will fail with index error” without executing it first. Pattern memory extracted patterns from revealed failures (e.g., “edge cases fail with empty arrays”) but failed to apply them (0% usage rate). Agents cannot infer failure modes without observations; clustering is pattern recognition (post-hoc), not causal modeling (predictive).

Interpretation of Transfer Failure: Contrary to initial expectation (clustering demonstrates “conceptual understanding” $\rightarrow$ should enable transfer), results show clustering and transfer are distinct capabilities. Post-hoc clustering (analyzing observed error messages) doesn’t require predicting unseen failures. This refines understanding from “strategic debugging = learning system extracting generalizable patterns” to “strategic debugging = feedback loop exploiting error message structure”. Framework measures iterative debugging with test execution, not autonomous code understanding.

6.2 Limitations

Mock Implementation Validity Threat: All experiments used mock agents (controlled `clustering_strength=0.5`, `fix_success_rate=0.6/0.7`) and synthetic datasets (balanced error types, controlled problem difficulty), not real GPT-4 API or Codeforces competitive programming problems. Mock validation establishes metric sensitivity (metrics *can* detect strategic behavior when engineered, validated via large effect size $d=3.07$), but ecological validity remains unverified — whether production agents exhibit clustering at coefficient $\bar{}$ 0.3 rates on real problems is unknown. Metrics capture structural properties (tests per modification, consecutive same-type error fixes) likely robust to dataset

source, but effect sizes ($d=3.07$, coefficient 1.909) observed in controlled conditions may be smaller or noisier with real agents facing genuine problem-solving complexity.

Why Acceptable: Controlled experiments validate framework’s discriminative power, establishing that metrics work in principle. Whether GPT-4 agents possess strategic debugging capability is separate empirical question — framework enables measurement when real-world deployment proceeds. Proof-of-concept establishes “can we measure strategic debugging?” (yes, via FIR/clustering/slope); production validation answers “do real agents exhibit it?” (future work FW3). This is standard PoC methodology: validate instrumentation on known signals before deploying to unknown phenomena.

Causal Chain Incomplete: Verified only Stages 1-2 (clustering $\rightarrow$ prioritization); Stage 3 (transfer) falsified, leaving mechanism incomplete. Cannot claim full understanding of strategic debugging when third predicted stage failed. However, partial two-stage mechanism provides coherent explanation for feedback-based debugging: agents cluster revealed errors (Stage 1, coefficient 1.909), prioritize high-impact fixes (Stage 2, 42.5% vs 19.8%), achieving efficiency within test-feedback loop. Failure of Stage 3 clarifies scope rather than invalidating contribution — framework evaluates iterative debugging with error access, not unsupervised pattern transfer.

Why Acceptable: Two-stage mechanism is self-contained and experimentally verified (h-m1/h-m2 both PASS with $p<0.05$). Falsified Stage 3 refines theoretical understanding (feedback loop vs learning system), demonstrates rigorous hypothesis testing (not all predictions confirmed), and identifies future research direction (alternative transfer designs FW7, pattern quality analysis FW1). We tested error-type clustering (syntax/runtime/logic/edge.case); alternative clustering dimensions (semantic similarity of error messages, code location, computational failure mode) may reveal additional mechanism variants worth exploring.

h-m2 Baseline Fairness Confound: While h-m2 validates that prioritization yields higher proportion of high-impact fixes (42.5% vs 19.8%), the experimental design confounds clustering strategy with fix success rate (proposed agent: 70% fix_success_rate + 60% cluster_bonus; baseline agent: 60% fix_success_rate + 0% cluster_bonus). The $2.15\times$ improvement may partially stem from unequal agent capabilities rather than clustering strategy alone. Future work should isolate prioritization effect by testing same-capability agents with/without clustering, controlling for fix success rate and cluster bonus mechanism.

Scope Limited to Multi-Test Scenarios: Framework applies when problems provide 15+ test cases with diverse error types (clustering requires error diversity, statistical power requires sufficient tests). Single-test benchmarks (HumanEval: 1-5 tests per problem) lack signal for clustering coefficient or held-out methodology. Applicability to low-test-count scenarios (≤ 10 tests) untested — clustering signal may degrade below detection threshold.

Why Acceptable: Scope boundaries principled and explicit. Multi-test competitive programming (Codeforces, LeetCode with 15-50 tests) and GitHub

issue solving (when test suites available) provide natural application domains. Framework addresses gap in existing benchmarks (HumanEval/MBPP measure single-attempt correctness, our metrics measure iterative debugging on multi-test problems) — complementary evaluation dimensions, not replacement.

Dataset Generalization Unknown: Mock synthetic problems (sum, max, array ops) with balanced 4-error-type distribution (25% each: syntax, runtime, logic, edge_case) may not reflect real Codeforces error distributions. Real competitive programming problems may exhibit skewed error types (80% logic errors, 5% syntax/runtime/edge_case), varying test quality (duplicates, incomplete coverage despite solve_count $\leq$ 1000 filtering), or domain-specific debugging patterns (algorithm complexity bugs require different clustering than implementation bugs). Effect sizes ($d=3.07$, coefficient 1.909) validated on controlled data; generalization to real datasets requires empirical confirmation (FW3).

Why Acceptable: Metrics designed to be domain-agnostic — fix-impact-ratio measures “tests passed per modification” regardless of problem topic, clustering coefficient uses permutation test controlling for problem-specific error distributions. Structural properties (fix patterns, consecutive error ordering) should remain measurable on real data, though noisier. Mock validation fulfills PoC requirement (MUST_WORK gate: metrics discriminate when strategic behavior present); real-world replication confirms ecological validity as natural follow-up.

6.3 Broader Impact

Positive Impacts: Framework establishes feasibility of benchmark design beyond HumanEval pass@k paradigm, measuring debugging efficiency alongside correctness. When validated on production systems, researchers can evaluate architectural improvements (memory modules, error-analysis prompts) via fix-impact-ratio changes, identifying which designs enhance strategic debugging. Practitioners deploying code-generation agents gain process metrics revealing *how* agents improve (clustering + prioritization vs trial-and-error), enabling informed architecture selection. Framework applicable to multi-test scenarios (competitive programming, GitHub issues if test suites exist), complementing existing correctness metrics.

Neutral/No Foreseeable Negative Impacts: Evaluation framework, not deployment system — measures debugging capability, doesn’t automate software development. No identified societal risks. Ecological validity unverified (mock agents) means production deployment requires real-world validation before high-stakes use.

Limitations on Claims: Framework reveals debugging patterns when multi-test feedback is available (15+ tests, diverse errors). Cannot evaluate single-attempt generation (HumanEval-style), code without test suites (style/maintainability assessment), or strategic debugging on ≤ 10 test problems (insufficient clustering signal). Transfer learning capability (Stage 3) not demonstrated — agents improve within revealed test feedback, don’t generalize to unseen tests autonomously.

6.4 Future Work

Immediate Extensions: (1) Validate with real GPT-4 API + Codeforces dataset (FW3) to confirm mock results generalize to production agents and real competitive programming problems. (2) Analyze pattern quality (FW1) — measure coverage (% revealed failures matched by patterns) and precision (% pattern predictions correct on held-out tests) to distinguish pattern extraction quality from fundamental transfer limitation. (3) Test oracle error patterns (FW2) — provide human-labeled high-quality patterns to agents, measure held-out slope; if ratio ≥ 1.5 with oracle patterns, transfer limitation is pattern quality; if ratio remains ≤ 1.5 , transfer is fundamentally information-limited (can't predict without test execution).

Longer-Term Vision: (1) Multi-capability benchmark suite evaluating debugging efficiency (fix-impact-ratio), test coverage generation, code efficiency (runtime, memory), and maintainability alongside correctness. (2) Extend to SWE-bench when test suites available — measure GitHub issue debugging efficiency, not just resolution rate. (3) Investigate alternative transfer mechanisms (few-shot learning FW7, meta-learning across problems, causal code-behavior modeling) to identify if any design enables pattern transfer where current approach failed. (4) Explore alternative clustering dimensions — semantic similarity of error messages (embeddings-based), code location (group errors by function/module), computational failure mode (timeout, memory error, wrong answer) — to test whether error-type clustering is one of many viable mechanism variants.

Open Questions: Can improved pattern extraction (better feature engineering, larger pattern training sets) enable transfer learning, or is predictive debugging without test execution fundamentally information-limited? Do production GPT-4 agents exhibit clustering behavior at rates (coefficient ≥ 0.3) matching mock validation, or is strategic debugging rare in real deployments? What clustering coefficient threshold discriminates strategic vs sequential debugging on real Codeforces problems with unknown error distributions?

7 Conclusion

We opened with a puzzle: an agent passes 95% of tests—but did it make 10 strategic root-cause fixes or 100 random modifications? Existing benchmarks can't tell. Our work in controlled settings shows that strategic debugging can be measured via fix-impact-ratio, a metric discriminating agents that target root causes (ratio 3.90: one fix resolves 3-4 test failures) from those using sequential trial-and-error (ratio 1.07: one fix per failure). Experiments reveal strategic debugging operates as a two-stage feedback loop — agents cluster errors by shared characteristics at $2\times$ random rate (coefficient 1.909, $p=0.001$), then prioritize fixes yielding $2.15\times$ higher proportion of high-impact modifications (42.5% vs 19.8%) — but transfer to unseen test cases fails (slope ratio 0.82 ≤ 1.5 , $p=0.504$). This clarifies that strategic debugging is a feedback loop effective within revealed

test execution, not a learning system enabling autonomous prediction without test execution.

7.1 Summary

We addressed the gap between outcome-focused metrics (pass@k) and process-based evaluation by introducing fix-impact-ratio, a framework measuring *how efficiently* agents debug through error feedback. Our main contributions are:

1. **Fix-impact-ratio framework with three execution-based metrics** — fix-impact-ratio (tests per modification), clustering coefficient (pattern recognition), held-out slope (generalization attempt) — validated with large effect size (Cohen’s $d=3.07$, $p=0.0001$) demonstrating strong discriminative power on controlled data.
2. **Experimental validation of two-stage mechanism** — agents cluster errors by type (coefficient 1.909 vs random 0.950, $p=0.001$), enabling prioritization of high-impact fixes (42.5% vs baseline 19.8%, $2.15\times$ improvement), explaining how strategic debugging achieves ratio 3.90 vs sequential 1.07. Transfer learning failed (slope ratio 0.82, $p=0.504$), refuting pattern-based generalization.
3. **Theoretical clarification** — strategic debugging is a *feedback loop* (error $\rightarrow$ cluster $\rightarrow$ prioritize $\rightarrow$ fix) effective when error messages guide clustering, not a *learning system* (observe $\rightarrow$ extract $\rightarrow$ predict) supporting transfer to unseen tests without execution. Framework measures iterative debugging with test access, complementing pass@k correctness metrics.

7.2 Future Directions

This work opens several research directions grounded in our experimental findings:

Validating Ecological Validity: Mock implementation (controlled `clustering_strength=0.5`) established metric sensitivity — fix-impact-ratio *can* detect strategic behavior when engineered ($d=3.07$). Whether production GPT-4 agents exhibit clustering at coefficient ≥ 0.3 rates on real Codeforces problems remains unverified. Immediate extension: replicate h-e1/h-m1/h-m2 with OpenAI API + curated Codeforces subset (solve_count ≤ 1000 , manual quality review) to confirm mock results generalize to real agents and competitive programming datasets.

Understanding Transfer Failure: Pattern memory extracted patterns from revealed test failures but achieved 0% usage rate on held-out tests. Two competing explanations: (1) pattern quality insufficient (low coverage/precision), or (2) transfer fundamentally information-limited (can’t predict without test execution). Test with oracle error patterns (human-labeled, high-quality) — if held-out slope ratio ≥ 1.5 with oracle patterns, transfer limitation is pattern extraction quality; if ratio remains ≤ 1.5 even with perfect patterns, predictive debugging without execution is fundamentally constrained.

Extending to Real-World Scenarios: Framework applies to multi-test scenarios (competitive programming, GitHub issue solving when test suites exist). SWE-bench evaluates issue resolution rates; fix-impact-ratio could measure debugging efficiency *within* resolution process, revealing *how* agents pass repository tests. Extension: analyze SWE-bench patches with test suites, compute fix-impact-ratio and clustering coefficient, compare agent strategies on real production codebases vs synthetic competitive programming.

Alternative Mechanism Variants: Current work tested error-type clustering (syntax/runtime/logic/edge_case); alternative clustering dimensions worth exploring: semantic similarity of error messages (embeddings-based), code location (group errors by function/module), computational failure mode (timeout, memory error, wrong answer). Each may reveal additional mechanism variants beyond error-type clustering, testing whether two-stage framework generalizes across clustering strategies.

Alternative Transfer Mechanisms: Current pattern memory (extract from revealed errors, apply to held-out tests) failed. Alternative designs worth exploring: few-shot learning (agent sees 2-3 examples of error type + fix, applies to similar held-out tests), meta-learning across problems (learn debugging strategy from Problem 1-40, apply to Problem 41-50), causal code-behavior modeling (simulate test execution without running tests). Each requires different agent capability than post-hoc clustering.

As code generation agents evolve from single-attempt generators to iterative problem-solvers, understanding *how* they improve becomes as critical as *whether* they succeed. Fix-impact-ratio establishes a measurement approach ready for real-world validation, proposing benchmark design beyond pass@k paradigm to measure debugging efficiency — a step toward comprehensive assessment of agentic capability in autonomous software development.

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